

Quantum Fault Tolerance Meets Toom's Contours

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1 Errors Clusters

Definition 1.1 (**Size()**-Function). Consider a (d, d') -Toric complex $\mathcal{G}$, and let U be a subset of vertices, $U \subset \mathcal{G}$. We define the *size* of U to be:

$$\mathbf{Size}(U) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

Furthermore, we extend the notion to binary assignment over the i th faces of a complex $z : \mathcal{F}^{(i)} \rightarrow \mathbb{F}_2$ by associating $\mathbf{Size}(z)$ with the size of the vertex subset supporting z .

Claim 1.2. Let $\mathbf{U} = \{U_1, U_2, \dots, U_k\}$ where $U_i \subset \mathbb{R}^d$, introduce an edge between U_i and U_j iff $U_i \cap U_j \neq \emptyset$. Assume that $\mathbf{U}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k U_i) \leq \sum_{i=1}^k \mathbf{Size}(U_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

Definition 1.3. We say that a Pauli operator P is an *X-type error* if it consists of a multiplication of single-qubit Pauli X and identities. Similarly, we say that P is a *Z-type error* if it consists of a multiplication of single-qubit Pauli Z and identities. For a binary assignment over the d' -faces, $z : \mathcal{F}^{(d')} \rightarrow \mathbb{F}_2$ we say that z is the *underline assignment* of P if z is the support of the non-trivial coordinates of P . We extend the size function to X/Z -types errors as follow:

1. If P is a X -type error, then $\mathbf{Size}(P) := \mathbf{Size}(z)$.

2. Else, namely P is a Z -type error, then $\mathbf{Size}(P) := \mathbf{Size}(\phi(z))$.

We will use the notation X/Z -type error to specify a Pauli that is either an X -type error or a Z -type error.

Claim 1.4. The following hold:

1. $\phi \circ H^{\otimes n}$ sends X -type errors to the same size Z -type errors, and vice versa.
2. $\mathbf{CX}^{\otimes n}$ and φ send X -type errors to the same size X -type errors. Similarly, they send Z -types errors to the same size Z -type errors.

Proof. Let z be a binary assignment over the d' -faces, $z : \mathcal{F}^{(d')} \rightarrow \mathbb{F}_2$, and let P be the X -type error which z is its underline assignment. Let $Q = \phi \circ H^{\otimes n} P$, and denote by z_Q the underline assignment of Q .

$$\begin{aligned} Q|\psi\rangle &= \phi \circ H^{\otimes n} P|\psi\rangle = \phi \circ H^{\otimes n} X_z|\psi\rangle \\ &= Z_{\phi(z)} \phi \circ H^{\otimes n} |\psi\rangle = Z_{\phi(z)} |\psi_2\rangle \end{aligned}$$

Where $|\psi_2\rangle$ is a codeword. Thus, Q is a Z -type error with underline assignment $z_Q = \phi(z)$, thus:

$$\mathbf{Size}(Q) = \mathbf{Size}(\phi(z_Q)) = \mathbf{Size}(\phi^2(z)) = \mathbf{Size}(z) = \mathbf{Size}(P)$$

For $\mathbf{CX}^{\otimes n}$, we consider two registers, R_1 and R_2 . The $\mathbf{CX}^{\otimes n}$ acts transversally on the registers by applying physical $\mathbf{CX}$ gates pairwise from R_1 to R_2 , and it's clear, by commutation relations, that it duplicates X -type errors from R_1 to their counterparts on R_2 . Similarly, $\mathbf{CX}^{\otimes n}$ duplicates Z -type errors from R_2 to their counterparts on R_1 .

Lastly, for φ , denote the permutation that induces φ by $\pi : S \rightarrow S$. Observe that for any subsets of vertices $U \subset \mathcal{G}$, we have:

$$\begin{aligned} \sum_{g \in S} L_g(\varphi(U)) &= \sum_{g \in S} L_{\pi(g)}(U) = \sum_{g \in \pi(S)} L_g(U) = \sum_{g \in S} L_g(U) \\ L_{d+1}(\varphi(U)) &= \max_{v \in \varphi(U)} - \sum_{g \in S} v_g = \max_{v \in U} - \sum_{g \in S} v_{\pi(g)} = L_{d+1}(U) \end{aligned}$$

Namely, the size function is invariant to φ -rotations, $\mathbf{Size}(\varphi(U)) = \mathbf{Size}(U)$. Hence:

$$\mathbf{Size}(\varphi(P)) = \mathbf{Size}(\varphi(X_z)) = \mathbf{Size}(X_{\varphi(z)}) = \mathbf{Size}(X_z) = \mathbf{Size}(P)$$

□

Corollary 1.5. Consider two registers R_1, R_2 , and let A be a logical operator which is realized by applications of $\psi \circ H^{\otimes n}$, $\mathbf{CX}^{\otimes n}$, and φ . Assuming that prior to an application of A , the total error over the two registers could be decomposed into X/Z -type errors $\mathbf{P} = P_1, \dots, P_l$, then the error after the application of A over the first (second) register can be decomposed into fewer than l X/Z -type errors $\mathbf{P}' = P'_1, \dots, P'_l$. There is an injective $h : \mathbf{P}' \rightarrow \mathbf{P}$ such that for any $P \in \mathbf{P}'$, we have:

$$\mathbf{Size}(P) = \mathbf{Size}(h(P))$$

Claim 1.6. Let R_1, R_2 be two registers, then any logical operator in $\mathcal{O}$ sends error cluster to an error cluster as follows:

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit). Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o). Let C be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, \dots, \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$E_o := \left\{ \{(v, t), (u, t')\} : \begin{array}{l} \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \end{array} \right\}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$). For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o . For each time t , denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are taken to be the vertices of G_o , namely tuples $(v, t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t , belongs to the $\partial\mathcal{D}_t$. Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where $A_{\mathcal{H}}$ are the edges in E_o restricted to $V_{\mathcal{H}}$ and $F_{\mathcal{H}}$ are defined to be the . Namely:

$$\begin{aligned} A_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\} \end{aligned}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 2.4 (Size Invariant Action). [COMMENT] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z , it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.5. The sweep rule is Size ξ -shrinking.

Proof.

□

[COMMENT] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time = $3t$, in that we apply an invariant action so.